

Week 1 - Introduction to AI Engineering

Concepts:

Overview of AI Engineering, AI Engineering vs AI Research, Importance of AI Engineering

Project:

Personal Project: Introduction to AI

Outcome:

Understand the basics of AI Engineering

Week 2 - Mathematics for AI

Concepts:

Linear Algebra, Calculus, Probability and Statistics

Project:

Personal Project: Math-based AI Model

Outcome:

Understand the mathematical foundations of AI

Week 3 - Programming Fundamentals

Concepts:

Python Programming, Data Structures and Algorithms, Object-Oriented Programming

Project:

Personal Project: AI-powered Chatbot

Outcome:

Develop programming skills for AI Engineering

Week 4 - Deep Learning Fundamentals

Concepts:

Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks

Project:

Personal Project: Image Classification Model

Outcome:

Understand the basics of Deep Learning

Week 5 - Natural Language Processing

Concepts:

Text Preprocessing, Tokenization, Named Entity Recognition

Project:

Personal Project: Sentiment Analysis Model

Outcome:

Understand the basics of NLP

Week 6 - Computer Vision

Concepts:

Image Processing, Object Detection, Image Segmentation

Project:

Personal Project: Object Detection Model

Outcome:

Understand the basics of Computer Vision

Week 7 - Reinforcement Learning

Concepts:

Markov Decision Processes, Q-learning, Policy Gradient Methods

Project:

Personal Project: Game Playing Model

Outcome:

Understand the basics of Reinforcement Learning

Week 8 - Transfer Learning and Fine-tuning

Concepts:

Transfer Learning, Fine-tuning, Domain Adaptation

Project:

Personal Project: Transfer Learning Model

Outcome:

Understand the concept of Transfer Learning and Fine-tuning

Week 9 - Explainability and Interpretability

Concepts:

Model Interpretability, Feature Importance, SHAP Values

Project:

Personal Project: Model Interpretability

Outcome:

Understand the concept of Explainability and Interpretability

Week 10 - Deploying AI Models

Concepts:

Model Deployment, API Design, Cloud Computing

Project:

Personal Project: Model Deployment

Outcome:

Understand the process of deploying AI models

Week 11 - Ethics and Bias in AI

Concepts:

AI Ethics, Bias in AI, Fairness in AI

Project:

Personal Project: Bias Detection Model

Outcome:

Understand the importance of Ethics and Bias in AI

Week 12 - Capstone Project

Concepts:

Project:

Personal Project: AI-powered Application

Outcome:

Apply AI Engineering skills to a real-world project
